Nevil Jasmin Mehta

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PROFESSIONAL SUMMARY

Backend-focused software engineer with experience building scalable systems and clean APIs using Python frameworks (FastAPI, Flask, Django), along with SQL/NoSQL databases and AWS. Passionate about exploring AI systems, particularly NLP, LLMs, and RAG pipelines, with hands-on experimentation and real-world implementations. Strong interest in system design, focusing on building efficient, scalable, and reliable architectures for real-world applications.

EDUCATION

Manchester Metropolitan University, UK Sept 2023–Dec 2024
MSc Computer Science & Engineering (**Distinction –** 76%)
Relevant Coursework: Focus on Algorithms & Data Structures, Cloud, System Design and Databases, and LLM, RAG concepts.

Indus University, India July 2018–June 2022
BTech Computer Science & Engineering (**First Class Honors –** 9.56/10 CGPI)
Relevant Coursework: Focus on Applied Mathematics, Operating Systems, Problem solving, Data Science, Cyber Security.

PROFESSIONAL EXPERIENCE

Full Stack Developer, Teknobu Solutions (UK) Mar 2025 – June 2025
- Developed a business website using React.js, Java backend, and MySQL, ensuring scalability and reliability.
- Integrated AWS cloud (Lambda, S3, Amplify, and Cognito) for deployment, authentication, and access control.
- Built a real-time voice interaction system for audio processing and response generation using cloud architecture.

Software Developer, InfiniteAi Technology Private Limited Jan 2023 – Aug 2023
- Delivered and maintained APIs using Django REST Framework, focusing on performance and system efficiency.
- Built responsive user interfaces using JavaScript, HTML, and CSS to enhance user experience and maintain the website.
- Optimized data queries and automated CAD workflows using Python, improving scalability and reducing manual effort.

Python Developer Intern, Rejoicehub Solutions Jan 2022 – May 2022
- Optimized backend APIs using Python frameworks, reducing response time by ~30% and improving scalability.
- Built backend systems with MongoDB and MySQL, implementing NLP-based data processing and text summarization.
- Utilized web scraping tools, deployed apps on Heroku, and built FastAPI projects including quiz and vehicle systems.

PROJECTS

AGSAQ – AI Sales Agent Platform, Freelance Project 2025
- Built an AI-powered chat platform with WhatsApp-style messaging and PDF summarization using LLM workflows.
- Integrated LangChain, FastAPI, and Gemini API for real-time interactions, designed for future voice features.

Job Market Intelligence Platform (Backend)
- Built a scalable backend system using FastAPI, PostgreSQL, Redis and Celery to collect, process & analyze job market data.
- Implemented background processing, data pipelines, analytics to identify trends, top companies & hiring patterns.

Plagiarism Detection for Large Language Models through Watermarking Method, Dissertation 2024
- Enhanced an AI content detection system using watermarking for LLMs, improving detection accuracy by 40%.
- Reduced detection failures by 15% through prompt engineering and model analysis, enhancing reliability.

Sentimental Analysis of Different Aspects, Industrial Project 2022
- Built a full-stack NLP application for sentiment and emotion analysis using real-time Twitter and YouTube data.
- Implemented ML and transformer models with Streamlit, enabling data processing, visualization & interactive insights.

SKILLS

Languages: Python, Java, C++, JavaScript, R
Frontend: React, Tailwind, HTML, CSS, Bootstrap, Streamlit
Backend: FastAPI, Flask, Django, RestAPI
Databases and DevOps: PostgreSQL, MySQL, MongoDB, Redis, AWS Cloud (S3, EC2, Lambda, IAM), Docker, Git
AI: NLP, LLM, RAG, LangChain, Web Scrapping (BeautifulSoup, Selenium)
Analytics: Pandas, Numpy, Matplotlib, Seaborn, Ploty